

ClipMixcure Manual

Changchang Xu

2026-03-18

Contents

1	Overview	1
2	Package data	2
3	Multiple imputation workflow	2
4	Fitting across imputations	2
5	CLIP inference	3
6	Rubin-style pooling	3
7	Meta-layer adapter	3
8	Notes	3
9	Session information	4

1 Overview

ClipMixcure supports multiply-imputed Weibull mixture cure modeling and combined likelihood profile (CLIP) inference. It is designed for settings where Rubin-style normal approximations may be unreliable and profile-based inference is preferred.

The main functions are:

- `mixcure.penal.mi()` for fitting mixture cure models across imputations
- `clip.mixcure()` for CLIP confidence intervals and p-values
- `pool.mixcure()` for Rubin-style pooled summaries
- `mixcure.clip.pdf()` for profile-density calculations used by CLIP
- `clipmixcure_fit()` as the adapter used by `Mixcuref.Meta`

2 Package data

The package includes one multiply-imputation example dataset:

- ANNbcBMdat5_miss

This dataset contains missing values and is intended for use with `mice`.

3 Multiple imputation workflow

```
library(ClipMixcure)
library(mice)
library(survival)

data(ANNbcBMdat5_miss)

imp20 <- mice(ANNbcBMdat5_miss, m = 20, maxit = 20, seed = 24415, print = FALSE)
data20 <- complete(imp20, action = "long", include = TRUE)

data20$TN <- factor(with(data20,
  ifelse((Her2 == "1" & ER == 2 & PR == 2), "2", "1")
))

data20$LuminalA <- factor(with(data20,
  ifelse((Her2 == "1" & (ER == 1 | PR == 1) & KI67 == 1), "2", "1")
))

data20$LuminalB <- factor(with(data20,
  ifelse((Her2 == "1" & (ER == 1 | PR == 1) & KI67 == 2), "2", "1")
))

imp20sub <- as.mids(data20)
```

4 Fitting across imputations

```
mixcure_all120 <- with(
  data = imp20sub,
  expr = mixcure.penal.mi(
    Surv(Time, CENS == 1) ~ Her2 + LuminalA + TN + MENS0 + TUMCT,
    init = c(2, rep(0.1, 5), -10, rep(0.1, 5), 1),
    pl = FALSE
  )
)

class(mixcure_all120)
length(mixcure_all120$analyses)
```

5 CLIP inference

```
out_clip <- clip.mixcure(  
  obj = mixcure_all20,  
  pl = FALSE  
)  
  
out_clip
```

Firth-type penalized estimation can be requested by setting `pl = TRUE` in both the fitting and CLIP steps.

```
mixcure_all20_penal <- with(  
  data = imp20sub,  
  expr = mixcure.penal.mi(  
    Surv(Time, CENS == 1) ~ Her2 + LuminalA + TN + MENO + TUMCT,  
    init = c(2, rep(0.1, 5), -10, rep(0.1, 5), 1),  
    pl = TRUE  
  )  
)  
  
out_clip_penal <- clip.mixcure(  
  obj = mixcure_all20_penal,  
  pl = TRUE  
)
```

6 Rubin-style pooling

```
pooled <- pool.mixcure(mixcure_all20)  
pooled
```

7 Meta-layer adapter

The `clipmixcure_fit()` adapter is used by `Mixcuref.Meta` to route engine-specific requests. It accepts an `mc_spec` object and engine-specific control arguments.

```
# spec is a model specification object created by Mixcuref.Meta  
# fit <- clipmixcure_fit(spec, control = list(  
#   method = "penal_mi",  
#   formula = Surv(Time, CENS == 1) ~ Her2 + LuminalA + TN + MENO + TUMCT,  
#   init = c(2, rep(0.1, 5), -10, rep(0.1, 5), 1),  
#   pl = FALSE  
# ))
```

8 Notes

- `mixcure.penal.mi()` is intended to be used inside `with(mids, ...)`.

- `clip.mixcure()` consumes the resulting `mira` object.
- CLIP intervals are especially useful when normal approximations are questionable.
- Derived subtype variables such as `TN` and `LuminalA` should be created before converting back to a `mids` object with `as.mids()`.

9 Session information

Table 1: Package Versions

Package	Version	Built	Remote
mice	3.19.0	R 4.5.2; x86_64-w64-mingw32; 2025-12-30 04:28:50 UTC; windows	
survival	3.8.3	R 4.5.0; x86_64-w64-mingw32; 2025-04-11 10:21:40 UTC; windows	
splines	4.5.0	R 4.5.0; x86_64-w64-mingw32; 2025-04-11 10:15:28 UTC; windows	
abind	1.4.8	R 4.5.0; ; 2025-04-08 00:42:55 UTC; windows	
R.utils	2.13.0	R 4.5.0; ; 2025-06-16 00:32:17 UTC; windows	